

Weighted trace-excess criterion and terminal anisotropic covariance reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-143 writes the owner-complete fixed-chart augmented action as

$$\mathcal{Y}_\pi(h) = \Phi_\pi(h) \oplus \sqrt{\frac{9}{10}} h \oplus \sqrt{\frac{3}{10}} |Z_{K,h}|^2 Z_{K,h}, \quad \mathcal{A}_\pi(h) = a_\pi + \frac{1}{2} \{ \|\mathcal{Y}_\pi(h)\|^2 - \|\mathcal{U}_\pi(h)\|^2 \}. \quad (1.1)$$

R-144 then gives two sufficient routes: a global common-output contraction with a bounded residual, or a positive reduced/source-low Hessian gap with an origin-force bound and an absolute anchor. Neither production input is registered.

This note takes a weaker, pointwise route. Define the complete trace excess

$$\mathcal{T}_\pi(h) = \frac{1}{2} \{ \|\mathcal{U}_\pi(h)\|^2 - \|\Phi_\pi(h)\|^2 \}. \quad (1.2)$$

The first result is an exact equivalence at the level needed by T-050: it is enough to bound $\mathcal{T}_\pi$ by the source and sextic budgets. No common linear map on a family of endpoints is required.

The second result is deliberately narrower. The terminal A1 total covariance has an exact scalar-principal plus positive anisotropic split. The complete spatial derivative trace of the anisotropic terminal remainder is trace class, has an explicit cutoff-independent lattice bound, and obeys a terminal trace-owner sextic estimate with coefficient 1/100 and certified constant at most 24 when the regulator bound is one.

The terminal lemma is not promoted through an arbitrary temporally faithful future-increment partition. R-129 proves that a total Fourier multiplier does not by itself intertwine each temporal covariance increment, and R-143 records the still missing output, reveal, collar, and low propagation. Thus the production temporal anisotropic owner, scalar principal sea, complete low/anchor owner, T-050, A13, Nelson estimate, removal, interacting measure, and Sector-A closure all remain open.

2. Exact direct trace-excess criterion

Put

$$X(h) = \mathbb{E} \sum_k \|h_k\|_2^2, \quad Y_6(h) = \mathbb{E} \|Z_{K,h}\|_6^6. \quad (2.1)$$

The norm convention in (1.1) gives

$$\frac{1}{2} \|\mathcal{Y}_\pi(h)\|^2 = \frac{1}{2} \|\Phi_\pi(h)\|^2 + \frac{9}{20} X(h) + \frac{3}{20} Y_6(h). \quad (2.2)$$

Consequently

$$\mathcal{A}_\pi(h) = a_\pi + \frac{9}{20} X(h) + \frac{3}{20} Y_6(h) - \mathcal{T}_\pi(h). \quad (2.3)$$

The controlled renormalised potential is obtained by subtracting the two one-use stabilisers already inserted in the augmented action:

$$\boxed{\mathbb{E}V_J^{\text{ren}}(Z_h) = a_\pi - \mathcal{T}_\pi(h).} \quad (2.4)$$

This cancellation is why the 9/20 and 3/20 coordinates do not shift the direct T-050 trace-excess thresholds.

Theorem 2.1 (direct weighted trace-excess criterion). Assume, uniformly in cutoff, admissible temporally faithful chart, floor, and control, that

$$a_\pi \geq -C_0, \quad \mathcal{T}_\pi(h) \leq \eta X(h) + \zeta Y_6(h) + B \quad (2.5)$$

for finite B, C_0 , with

$$\boxed{\eta < \frac{5}{11} = \frac{1}{2p}, \quad \zeta < \frac{27}{100} = \frac{\gamma}{6}, \quad p = \frac{11}{10}.} \quad (2.6)$$

Then T-050 holds with $\epsilon_v = \eta$, $\epsilon_6 = \zeta$, and $C = B + C_0$.

Proof. Equations (2.4)–(2.5) give directly

$$\mathbb{E}V_J^{\text{ren}}(Z_h) \geq -\eta X(h) - \zeta Y_6(h) - (B + C_0). \quad (2.7)$$

The strict inequalities (2.6) are exactly the two canonical T-050 acceptance conditions. $\square$

This theorem is weaker than demanding one common contraction $\mathcal{U} = C\mathcal{Y}$ on a whole endpoint family. For example, take two paid endpoint vectors with Gram $G_{\text{paid}} = I_2$ and an adverse Gram

$$G_U = \frac{9}{16} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (2.8)$$

Each diagonal endpoint has adverse squared norm $9/16 < 1$, but G_U has largest eigenvalue $9/8 > 1$, and $G_{\text{paid}} - G_U$ has eigenvalue $-1/8$. Thus endpointwise weighted control may hold although the stronger mixed-Gram contraction fails. The fixture does not prove the production inequality (2.5); it proves only that failure of a common contraction is not a logical obstruction to it.

2.1 R-129 acceptance-window audit

R-129 requested

$$\eta < \frac{9}{20}, \quad \zeta < \frac{3}{20}. \quad (2.9)$$

That result is valid: (2.9) makes the augmented action in (2.3) positive up to its anchor. It is, however, stronger than T-050 requires. The direct renormalised-potential criterion is (2.6), with exact additional headroom

$$\frac{5}{11} - \frac{9}{20} = \frac{1}{220}, \quad \frac{27}{100} - \frac{3}{20} = \frac{3}{25}. \quad (2.10)$$

Accordingly the normalized feature comparison

$$G_U - G_\Phi \preceq \alpha_v G_{\sqrt{9/10}h} + \alpha_6 G_{\sqrt{3/10}|Z|^2Z} \quad (2.11)$$

would be sufficient pointwise when

$$\alpha_v < \frac{100}{99}, \quad \alpha_6 < \frac{9}{5}. \quad (2.12)$$

The audit is a nonsharp-scope correction, not a refutation of R-129.

3. Terminal total-covariance anisotropy

The A1 family masses and lock give the exact mass matrix

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (3.1)$$

Exact Sturm chains isolate its three eigenvalues in

$$(28, 29)/1000, \quad (165, 166)/1000, \quad (206, 207)/1000. \quad (3.2)$$

Writing them as $\mu_1 < \mu_2 < \mu_3$, these intervals imply

$$\Delta_\Sigma = (\mu_3 - \mu_1) + (\mu_3 - \mu_2) \leq \frac{221}{1000}. \quad (3.3)$$

Let $|\chi_J(k)| \leq M_\chi$. For the terminal total covariance put

$$C_J(k) = \chi_J(k)^2 (a(k)I + M)^{-1} = \gamma_{0,J}(k)I + R_J(k), \quad (3.4)$$

where

$$\gamma_{0,J}(k) = \frac{\chi_J(k)^2}{a(k) + \mu_3}, \quad R_J(k) \succeq 0. \quad (3.5)$$

In the mass eigenbasis, the resolvent identity gives

$$\mathrm{Tr}_{\mathbb{C}^3} R_J(k) \leq M_\chi^2 \frac{\Delta_\Sigma}{a(k)^2}. \quad (3.6)$$

The pinned kinetic polynomial is

$$a(k) = r + Z|k|^2 + Y|k|^4, \quad Y = 1, \quad (3.7)$$

and the exact A1 inputs satisfy

$$r - \frac{Z^2}{4Y} = \frac{26000000000947494031}{10^{20}} > \frac{1}{4}, \quad (3.8)$$

as well as

$$a(k) \geq (|k|^2 - q_0^2)^2. \quad (3.9)$$

The check $Y = 1$ is load bearing in (3.9).

4. Complete spatial lattice trace bound

On the $L = 16$ torus write $k = (\pi/8)n$ and $m = \|n\|_\infty$. Direct summation gives

$$\sum_{\|n\|_\infty \leq r} |n|^2 = r(r+1)(2r+1)^3, \quad (4.1)$$

hence the exact shell identity

$$\boxed{\sum_{\|n\|_\infty = m} |n|^2 = 40m^4 + 14m^2.} \quad (4.2)$$

The radius-two cube has squared-radius sum 750.

Use the rational enclosure $3 < \pi < 22/7$. On the low cube, $a(k) > 1/4$, so (3.3), (3.6), and (4.1) give the unnormalised complex Fourier-sum bound

$$G_{R,\text{low}}^{\mathbb{C}} \leq M_{\chi}^2 \frac{221}{1000} 16\,750 \left(\frac{11}{28}\right)^2 = M_{\chi}^2 \frac{80223}{196}. \quad (4.3)$$

For $m \geq 3$, the exact A1 value satisfies $q_0/(\pi/8) < 2$. Therefore

$$a(k) \geq c^2 |k|^4, \quad c = 1 - \frac{4}{32} = \frac{5}{9}. \quad (4.4)$$

The integral test gives

$$\sum_{m \geq 3} m^{-4} \leq \frac{2}{81}, \quad \sum_{m \geq 3} m^{-6} \leq \frac{8}{3645}. \quad (4.5)$$

Combining (3.3), (3.6), and (4.2)–(4.5) yields

$$G_{R,\text{tail}}^{\mathbb{C}} \leq M_{\chi}^2 \frac{26881294336}{31640625}. \quad (4.6)$$

Thus

$$G_R^{\mathbb{C}} := \sum_k |k|^2 \text{Tr}_{\mathbb{C}^3} R_J(k) \leq M_{\chi}^2 \frac{7807039549231}{6201562500}. \quad (4.7)$$

Equation (4.7) is an unnormalised complex Fourier sum, not yet the physical pointwise six-real trace. The A6/A7 convention realifies the complex covariance with a factor two and uses the $1/V$ Fourier normalization. Since $V = 16^3 = 4096$,

$$g_R^{(6)} = \frac{2}{V} G_R^{\mathbb{C}} \leq M_{\chi}^2 \frac{7807039549231}{12700800000000}. \quad (4.8)$$

The factor $2/V$ is an explicit normalization firewall. The lattice extension is a uniform upper bound conditional on the registered A1 symbol and regulator hypotheses; A1 itself remains a finite-grid authority.

5. Terminal anisotropic trace-owner lemma

This section states exactly what (4.8) proves and no more. Let Γ_R be a translation-invariant terminal derivative covariance with pointwise six-real trace at most $g_R^{(6)}$. Let P be an orthogonal output projection and let the registered Pauli–Fierz coordinate obey

$$0 \preceq C_6(z)^* P C_6(z) \preceq \beta_{\text{op}} |z|^2 \mathbf{I}. \quad (5.1)$$

From the A1 coefficients,

$$\beta_{\text{op}} = \frac{169500000000}{4000000000001} < \frac{339}{8000}. \quad (5.2)$$

The coefficient Gram determinant is positive, and (5.1) follows from the Pauli operator norms and the triangle bound on the three coefficient rows.

For a full terminal random field Z_h , define the abstract terminal anisotropic trace owner

$$T_{\text{an}}(Z_h) = \frac{1}{2} \mathbb{E} \int \text{Tr} \{ P C_6(Z_h) \Gamma_R C_6(Z_h)^* P \} dx. \quad (5.3)$$

Theorem 5.1 (terminal anisotropic sextic payment). Under (4.8) and (5.1),

$$T_{\text{an}}(Z_h) \leq A_R Y_6(h)^{1/3}, \quad A_R = \frac{\beta_{\text{op}}}{2} g_R^{(6)} V^{2/3}. \quad (5.4)$$

7. Independent Branch-B audit

R-141 supplies the exact finite-chart action and projected force. A conditional Branch-B theorem would follow from a uniform positive Hessian gap on one declared reduced source/low action, a uniform origin-force bound, and a uniform absolute anchor. None is currently a production input.

The low coordinate cannot simply be minimized as a free stationary variable. The current conditional low and returned low are prescribed, distinct owner functionals. If low is a nonlinear graph $\Lambda(h)$, the pullback Hessian contains the connection term

$$D\Lambda D^2\Lambda, \quad (7.1)$$

not only the Schur complement of an ambient Hessian. A stationary reduction is legal only after an attained C^2 low minimizer with the same action, force, and anchor is constructed. Branch B therefore remains data-gated; it is not an escape from the scalar signed trace-excess problem.

8. Proof-route and evidence map

Route	Verdict	Reusable consequence
Global $\mathcal{U} = C\mathcal{Y}$, $\ C\ < 1$	conditional	Sufficient by R-143/R-144, but stronger than the pointwise T-050 burden.
Bounded affine residual	conditional	R-144 permits a uniformly bounded returned-low residual; no production bound is registered.
Direct trace excess (2.5)	advanced	Exact T-050 criterion with $\eta < 5/11$, $\zeta < 27/100$.
R-129 window	valid, nonsharp	9/20, 3/20 guarantees augmented-action positivity; it is not the maximal T-050 window.
Terminal anisotropic co-variance	advanced, scoped	Exact trace-class bound and terminal sextic payment; no temporal owner transfer.
Scalar principal absolute trace	failed	Shell contribution tends to 24; signed cancellation is load bearing.
Stationary-low Schur reduction	unavailable	Current low is a prescribed graph; connection terms or an attained C^2 minimizer are required.
Reduced Hes-sian/force/anchor	conditional	Exact R-144 implication, but all three production-uniform inputs are absent.

This map separates a proved theorem, a conditional implication, a failed method, and missing data. It must not be read as eight independent claims; they are subroutes of the single A13/T-050 burden.

9. Devil's-advocate review

- Objection: The trace-excess thresholds should be shifted by 9/20 and 3/20. DISMISSED.** Those terms belong to the augmented action and are subtracted in (2.4). T-050 concerns $V^{\text{ren}} = a_\pi - \mathcal{T}_\pi$, so the direct thresholds are 5/11 and 27/100.
- Objection: R-129 was wrong. DISMISSED.** Its window is a valid stronger sufficient window which makes the augmented action positive. This note corrects only its use as the purported full T-050 acceptance range.
- Objection: The raw lattice sum is already the physical trace. UPHOLD against the shortcut.** The six-real pointwise trace is $(2/V)G_R^{\text{C}}$. Equations (4.7)–(4.8) keep realification and Fourier normalization explicit.

4. **Objection: The terminal covariance split propagates through every temporal reveal. UPHELD.** R-129 gives a finite-dimensional noncommuting increment fixture, and R-143 lists the missing owner data. The production-temporal transfer is false as an inference and remains open as a theorem.
5. **Objection: The rounded decimal 23.432... is a certified upper bound. UPHELD against that presentation.** Decimal square roots round. The certificate instead proves the rational squared inequality (5.10) and reports the integer ceiling 24.
6. **Objection: The kinetic square bound follows merely from $Y > 0$. UPHELD.** The displayed comparison with coefficient one uses the pinned $Y = 1$. Both executable paths assert that exact value before using (3.9).
7. **Objection: The anisotropic estimate closes T-050. DISMISSED.** It is terminal-only. The scalar principal sea, production temporal transfer, complete low/anchor, and the full trace-excess estimate remain open.
8. **Objection: A positive reduced Hessian removes the need for an anchor. DISMISSED.** Convexity controls variation, not absolute normalization. Branch B still needs the same reduced action's origin force and anchor uniformly.

10. Result footer

The primary certificate uses exact rational arithmetic and SymPy Sturm chains. The independent certificate uses only the Python standard library, implements polynomial division and Sturm counting directly, and enumerates the integer shells. They independently reconstruct the A1 mass, kinetic and coefficient inputs, regulator scaling, Fourier normalization, lattice sum, Young ceiling, and T-050 thresholds, then compare every exact output.

Dependencies: A1, A7, GATES, R-104, R-129, R-130, R-141–R-144.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_weighted_trace_excess_anisotropic_covariance_
sextic_reduction_verify.py
```

Expected output: integrated PASS after primary 71/71 and non-importing independent 90/90, with deterministic PDF, authority, scope, and public-surface checks.

Result ID: A13-CLASSII-WEIGHTED-TRACE-EXCESS-ANISOTROPIC-
COVARIANCE-SEXTIC-REDUCTION.

Ledger ID: R-145.

Precise statement: direct trace excess below 5/11 and 27/100 implies
T-050 without a common contraction. The terminal anisotropic
covariance is trace class and its abstract terminal owner is paid
by $\text{zeta}_{\text{an}}=1/100$ and constant 24 when $M_{\text{chi}} \leq 1$.

Scope: terminal total covariance; uniform lattice majorant conditional
on the A1 symbol/regulator; no temporal-owner transfer.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Audit: AUDIT-2026-08-02-A13-R129-TRACE-EXCESS-ACCEPTANCE-WINDOW.

Falsification gate: any failed hash, mass/kinetic/shell calculation,
2/V normalization, coefficient, Young, scope, or release assertion.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no production temporal owner, scalar principal estimate,
complete low/anchor, T-050, A13, Nelson, measure, or Sector-A closure.

Proof complete: false. T-050 closed: false. Sector A closed: false.

Next required action: transfer terminal anisotropy legally; then prove the
signed scalar sea and complete low/anchor owner.